

CLASS 9th

MATHS BASIC NOTES

1. Number Systems

- **Rational Numbers:** Can be expressed as $\frac{p}{q}$, $q \neq 0$.
- **Irrational Numbers:** Cannot be expressed as $\frac{p}{q}$, e.g., $\sqrt{2}$, π .
- **Real Numbers:** All rational and irrational numbers.
- **Operations on Real Numbers:** Closure, commutative, associative, distributive laws.
- **Laws of Exponents:**

$$a^m \cdot a^n = a^{m+n}, \frac{a^m}{a^n} = a^{m-n}, (a^m)^n = a^{mn}$$

2. Algebra

- **Polynomials:** Expressions like $ax^2 + bx + c$.
- **Degree of Polynomial:** Highest power of the variable.
- **Factorization:**
 - Common factor: $ax + ay = a(x + y)$
 - Difference of squares: $a^2 - b^2 = (a - b)(a + b)$
 - Perfect square trinomial: $a^2 + 2ab + b^2 = (a + b)^2$
- **Algebraic Identities:**

$$\begin{aligned}(a + b)^2 &= a^2 + 2ab + b^2 \\(a - b)^2 &= a^2 - 2ab + b^2 \\a^3 + b^3 &= (a + b)(a^2 - ab + b^2) \\a^3 - b^3 &= (a - b)(a^2 + ab + b^2)\end{aligned}$$

3. Geometry

- **Triangles:**
 - Types: Equilateral, Isosceles, Scalene
 - **Pythagoras Theorem:** $a^2 + b^2 = c^2$ for right triangles.
- **Quadrilaterals:** Properties of parallelogram, rectangle, square, rhombus.

- **Circles:**
 - Chord, tangent, secant definitions
 - Angle subtended by chord at center = twice the angle at any point on the circle
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4. Coordinate Geometry

- **Distance Formula:**

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- **Midpoint Formula:**

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

5. Mensuration

- **Surface Area & Volume:**

- Cube: $SA = 6a^2, V = a^3$
 - Cuboid: $SA = 2(lb + bh + hl), V = l \cdot b \cdot h$
 - Sphere: $SA = 4\pi r^2, V = \frac{4}{3}\pi r^3$
 - Cylinder: $SA = 2\pi r(h + r), V = \pi r^2 h$
 - Cone: $SA = \pi r(l + r), V = \frac{1}{3}\pi r^2 h$
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6. Statistics

- **Mean:** $\bar{x} = \frac{\sum x_i}{n}$
 - **Median:** Middle value when data arranged in order.
 - **Mode:** Value that occurs most frequently.
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7. Probability

- Probability of an event E :

$$P(E) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}}$$

8. Linear Equations

- Standard form: $ax + by + c = 0$
 - Solving using substitution, elimination, or graphical method.
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9. Quadrilaterals & Polygons

- Sum of interior angles of n-sided polygon: $(n - 2) \cdot 180^\circ$
 - Sum of exterior angles = 360°
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10. Surface Area & Volumes (3D Shapes)

- Learn formulas for cube, cuboid, sphere, cylinder, cone, and frustum.
- Volume of frustum of cone:

$$V = \frac{1}{3}\pi h(R^2 + r^2 + Rr)$$